# 0. How to use this document

This file captures the OSU IRB and IEEE publication requirements that govern the sEMG/IMU study. It is the **reference context** when drafting `IRB_Packet_OSU_sEMG_IMU_Athletic_Practice.md` and the ethics sections of any **future IEEE-targeted manuscript** that draws on data collected under this protocol. Direct quotes from authoritative sources are marked with `>`. Items marked **TBD** require user/PI input before submission.

**Scope clarification.** This IRB packet is being prepared to support **future IEEE biomedical submissions**. Any prior or in-flight engineering write-ups produced from pre-IRB engineering test data are out of scope of this protocol and are not "research analysis, conference submission, or publication" as those terms are used in this document; only data collected after IRB approval will be used to support IEEE-targeted research publications.

When IEEE policy and OSU policy collide, OSU policy controls study conduct (it has legal force); IEEE policy controls only what must be in the manuscript.

# Part 1 — OSU IRB Requirements

## 1.1 Submission mechanics

Source: OSU IRB main page · New Applications

| Item | Value |
|---|---|
| Portal | Cayuse HE — `oregon-state.app.cayuse.com` |
| Authentication | OSU SSO (ONID) |
| PI eligibility | Must hold an OSU appointment and act within its scope |
| Application content | Cayuse form + study instruments + recruitment + consent/assent + data security plan |
| Pre-submission tip | Confirm "Review-ready" status before submission to shorten turnaround |

## 1.2 Governing regulations and policy documents

Source: OSU HRPP Policy and Procedure Manual

- **U.S. Common Rule** — 45 CFR 46 (revised 2018), governs all federally-conformant U.S. human-subjects research
- **45 CFR 46 Subpart D** — Additional Protections for Children
- **OSU HRPP Policy and Procedure Manual v4.6.1** — operative OSU document, governed by 45 CFR 46.103

## 1.3 CITI training (mandatory before submission)

Source: OSU IRB — Education and Advising · CITI Program

- All study team members complete **Group 1 or Group 2 — Human Subjects Protection** in CITI
- **Important:** Responsible Conduct of Research (RCR) training is **not** a substitute for the required Human Subjects Protection course at OSU. Group 1 or Group 2 is required.
- Renewed every **3 years**
- Email in CITI must match Banner email so records link to Cayuse HE
- Non-OSU team members: complete OSU CITI (with OSU affiliation added) **or** submit equivalent training from home institution acceptable to the IRB
- GCP training (Group 5/6/7) only required for FDA-regulated or NIH clinical trials — **not applicable** to this minimal-risk pilot

## 1.4 Review categories (timelines published by OSU)

Source: OSU IRB — Submission Information · 45 CFR 46.110 (eCFR)

| Category | Typical timeline | Applies when |
|---|---|---|
| Exempt | 3–4 weeks | Specific 45 CFR 46.104 categories — generally **not** for research with minors involving sensors |

| Category | Typical timeline | Applies when |
|---|---|---|
| **Expedited** | **6–8 weeks** | **Anticipated for this study** — minimal-risk research involving noninvasive physiological sensor recording and research on individual characteristics, behavior, and movement during ordinary athletic practice (tennis practice for one participant; running at a neighborhood fitness center for the other). **OSU determines the applicable expedited category** under 45 CFR 46.110; the request will not pre-assign a category. |
| Full Board | 12 weeks | Greater than minimal risk, vulnerable populations beyond minor-only, etc. |

OSU determines the category from the application; the applicant requests but does not assign.

## 1.5 Required submission components

Source: OSU IRB — New Applications · Consent and Assent Templates

1. Cayuse application form (study team, IRB record, etc.)
2. Protocol document
3. Consent form(s) using OSU template
4. Assent form using OSU template
5. Recruitment materials
6. Test instruments (subjective scales, questionnaires)
7. Data security plan (especially if Level 3 data)
8. Conflict of interest disclosure for each team member with disclosable interests

**OSU current template choices for this study:**

- Parental permission form: adapt OSU's **"Consent Form template for non-exempt studies"** (covers Minimal Risk, FDA-regulated, More Than Minimal Risk) — **do not** use the exempt-study template, since this protocol is non-exempt expedited.
- Minor assent form: use OSU's current **"Assent Form"** template.
- Procedure/population-specific language: pull from OSU's **"Template language for procedures and populations"** document.
- **Do not** reuse previously approved consent/assent forms from other studies; OSU explicitly asks researchers to use the current New-Common-Rule templates for all new submissions.

## 1.6 Conflict of interest

Source: OSU Office of Research Integrity — Conflict of Interest

- COI disclosures filed via **Beaver Research Hub**
- COI Office: A312 Kerr Administration, **541-737-2762**
- **Re-verification note:** the OSU COI page returned partial content during automated fetch; before final IRB submission, the PI should confirm the current Beaver Research Hub workflow and any familial-relationship disclosure templates directly with the OSU COI Office.
- OSU public pages do not enumerate familial-relationship-with-participant scenarios; this study will propose a **written COI Management Plan** as part of the IRB packet (Doc 6)
- For this study, both minor participants are sons of team members → **symmetric undue-influence problem** → mitigation requires an Independent Assent Administrator (Doc 8)

## 1.7 Data security and privacy

Source: OSU IRB — New Applications (Data security plan)

- OSU classifies data into Levels 1–3 (Public / Confidential / High)
- **Level 3** data security plans must be reviewed by the OSU Information Security Office before research starts
- This study collects raw sEMG, IMU, and 0–10 subjective ratings tied to a single named minor each, with no PHI/PII beyond name and code → **likely Level 2 (Confidential)**, not Level 3
- **N=2 indirect-identifiability warning:** because the cohort is two people and both participants may be identifiable through manuscript authorship and study context, the protocol treats raw and coded physiological data as **potentially identifiable even after removal of direct identifiers**. Public sharing of raw participant-level data is **not** promised; data-availability statements (IEEE) reflect this constraint.
- **Level 2 cloud storage caveat:** per OSU guidance, if coded or identifiable Level 2 data are stored on a cloud-based server, the data-security plan must be reviewed by the OSU Information Security Office **unless** the cloud server is licensed by OSU. To avoid triggering additional ISO review, this study will use **OSU-licensed storage or encrypted local storage** with controlled access.
- No video, audio, heart-rate, $SpO_2$, or other vital-sign data
- Coded storage with the identity-key file held separately on encrypted local drive
- Routine encrypted backup; patched OS; firewall; access limited to named study team only

## 1.8 Subpart D — Children in research

Source: 45 CFR 46 Subpart D (eCFR)

- **Risk classification:** 45 CFR 46.404 — *Research not involving greater than minimal risk*
- **Permission requirement (§46.408(b)):**
  - For research approvable under §46.404 (this study) or §46.405, the IRB **may** find that permission of **one parent** is sufficient.
  - For research under §46.406 or §46.407, permission of **both parents** is required unless one parent is deceased, unknown, incompetent, not reasonably available, or only one has legal responsibility for the child.
  - **Study-specific stance:** because both minor participants are children of individuals affiliated with the study team, the protocol will seek written permission from **both parents when feasible** as an additional protection against perceived undue influence — even though the regulation does not require it for §46.404 work. If only one parent signs, the application must document the regulatory basis and the IRB's accepted justification.
- **Assent (§46.408(a)):** Required from the child where in the IRB's judgment the children are capable of providing assent. Both Ryan (17) and Theodore (15) are capable of providing meaningful written assent.
- **Exemption pathway not used:** although certain Common Rule exemptions can apply to research with children, this study should **not** rely on exempt status because it involves minors, coded/identifiable physiological sensor data, and research procedures during physical activity. The application is framed as **non-exempt minimal-risk research under §46.404**, anticipated eligible for **expedited** review at OSU's discretion.
- **18th-birthday transition:** Re-consent as adult required if data continues to be collected after the participant turns 18; previously-collected data retained under parental permission stays usable

## 1.9 Off-site / multi-state data collection

Source: OSU HRPP Policy and Procedure Manual

- OSU IRB has jurisdiction via the PI's institutional affiliation regardless of where data is collected
- Protocol must **describe** non-OSU sites (residential court for tennis, neighborhood fitness center for running; weather/safety/equipment procedures; supervising adult)
- Multi-state participation (Ryan in CA, Theodore in OR) is allowed under OSU IRB but must be explicitly described
- No school-based-research clearance needed because data collection is at a residential tennis court (Ryan) and a neighborhood fitness center (Theodore), not at Westview HS or Crescent Valley HS

## 1.10 Continuing review and amendments

Source: OSU IRB — Submission Information

- Cayuse HE manages amendments, continuing review, and adverse-event reporting
- Adverse events (skin reaction, injury, distress) reported through Cayuse per HRPP timelines
- Annual continuing review for non-exempt protocols (timeline depends on review category and approval letter)

# Part 2 — IEEE Publication Requirements

## 2.1 Verbatim policy text (IEEE Author Center, Submission and Peer Review Policies)

Source: IEEE — Submission and Peer Review Policies

**IRB statement requirement** (direct quote):

> "the research was performed under the oversight of an institutional review board or equivalent local/regional body, including the official name of the IRB/ethics committee"

(or, if not reviewed, an explanation of why such review wasn't conducted)

**Consent statement requirement** (direct quote):

> "shall also report that consent from the human subjects in the research was obtained or explain why consent was not obtained"

## 2.2 Authorship criteria (IEEE Ethical Requirements)

Source: IEEE — Ethical Requirements · ICMJE Authorship Criteria

Authors must meet **all three** criteria, including:

> "made a significant intellectual contribution to the theoretical development, system or experimental design, prototype development"

(plus drafting/revising and final approval / accountability — standard ICMJE-style triad).

Contributors not meeting all three criteria belong in **Acknowledgments**, not the author list.

**Implication for this study:**

- Ryan Wang — qualifies (system design participation, primary data collection, analysis pipeline, drafting) — **lead student author**
- Theodore (Teddy) Di — qualifies via second-site data collection, cross-site reproducibility analysis, and manuscript review — **defined co-author role**
- Dr. YanMing Di — qualifies via study design, statistical oversight, and supervisory role — **PI / corresponding author**
- Xiaodong Wang (Technical Assistant) — does **not** qualify under IEEE criteria because his role is engineering support and on-court supervision, not the manuscript's intellectual content → **Acknowledgments only, not an author**

## 2.3 Conflict of interest

Source: IEEE — Submission and Peer Review Policies (COI section)

IEEE definition (paraphrased): any situation, transaction, or relationship in which someone's decisions or actions could materially affect that individual's professional, personal, financial, or business concerns. Reviewers and editors with conflicts must recuse.

**Implication:** the Author COI declaration in the manuscript must disclose:

- Dr. Di is the parent of co-author/participant Theodore
- Xiaodong Wang (in Acknowledgments) is the parent of co-author/participant Ryan
- Independent Assent Administrator details

## 2.4 Plagiarism, originality, prior publication

Source: IEEE — Ethical Requirements · Submission and Peer Review Policies

- All articles screened for plagiarism before publication
- Submission must be original, not previously published, not under simultaneous review elsewhere
- Conference→journal evolution allowed if the journal version "contains substantially more technical information"; all prior versions must be referenced
- Self-plagiarism (recycled text from author's own prior work) must be marked, referenced, and explained

## 2.5 Data integrity

Source: IEEE — Ethical Requirements (Data Reporting)

No fabrication, no falsification, no image manipulation that alters scientific meaning.

## 2.6 AI-generated content disclosure

Source: IEEE — Submission and Peer Review Policies (AI-Generated Content)

> Use of AI-generated content (text, figures, images, code) "shall be disclosed in the acknowledgments section."

Disclosure must identify: the AI system used, the specific sections involved, and a brief usage explanation. Editing/grammar tools are exempt but disclosure is recommended.

## 2.7 Preprint posting

Source: IEEE — Post-Publication Policies · Submission and Peer Review Policies

Permitted with required boilerplate before acceptance:

> "This work has been submitted to the IEEE for possible publication. Copyright may be transferred without notice, after which this version may no longer be accessible."

## 2.8 Biomedical-venue norms (not core IEEE policy, but expected by EMBC / JBHI / JTEHM / TBME / Sensors)

These are **not** in IEEE's core ethics text but are standard practice and expected by reviewers in biomedical venues:

| Item | Norm | Reference |
|---|---|---|
| Declaration of Helsinki | Conformance statement in Methods (e.g., "The study was conducted in accordance with the Declaration of Helsinki and approved by [IRB name], protocol number [#]") | WMA Declaration of Helsinki |
| Author contributions | CRediT taxonomy or narrative section listing each author's role | CRediT taxonomy |
| Data availability statement | Required by some IEEE biomedical journals; for this study expect: "Raw data are not publicly available because the IRB-approved protocol does not authorize public sharing of single-participant identifiable physiological data; de-identified summary metrics are available from the corresponding author on reasonable request." | Verify per venue |

| Item | Norm | Reference |
|---|---|---|
| Funding statement | Required even if "This work received no external funding." | Standard |
| ORCID | Compulsory for IEEE Sensors Journal; recommended elsewhere | ORCID |

## 2.9 Venue-specific requirements (verify when target chosen)

The exact submission target affects which biomedical norms above are mandatory vs. recommended. **Re-verify the venue's author guide before submission.** Likely targets:

- IEEE EMBC (Engineering in Medicine and Biology Conference) — annual; check the current-year `authors-presenters` page
- IEEE Journal of Biomedical and Health Informatics (JBHI)
- IEEE Journal of Translational Engineering in Health and Medicine (JTEHM)
- IEEE Transactions on Biomedical Engineering (TBME)
- IEEE Sensors Journal
- IEEE Sensors Letters

# Part 3 — Compliance Mapping for This Study

| Requirement | Source | Where it lives in our work |
|---|---|---|
| IRB approval reference | IEEE 2.1 | Manuscript Methods ("approved by Oregon State University HRPP, protocol #[TBD]"); IRB Packet Doc 1 |
| Consent statement | IEEE 2.1 | Manuscript Methods (parental permission + minor assent obtained); IRB Packet Docs 2 & 3 |
| Helsinki conformance | IEEE biomedical norm 2.8 | Manuscript Methods; IRB Packet Doc 1 |
| Subpart D handling | OSU 1.8 | IRB Packet Doc 1 (procedures), Doc 2 (parental permission for both kids), Doc 3 (assent for both kids) |

| Requirement | Source | Where it lives in our work |
|---|---|---|
| Two-parent permission (study-specific stance) | OSU/Subpart D §46.408(b) — IRB may allow one parent for §46.404 work, but we elect both | Both Ryan's and Theodore's parents sign permission when feasible; IRB Packet Doc 2A & 2B |
| 18th-birthday re-consent (Ryan ~Sep 2026) | OSU 1.8 | IRB Packet Doc 1 §6 + amendment plan |
| 18th-birthday provision (Theodore ~Sep 2028) | OSU 1.8 | IRB Packet Doc 1 §6 |
| CITI training | OSU 1.3 | IRB Packet Cover Letter + Appx D (CITI tracker) |
| COI Management Plan | OSU 1.6 + IEEE 2.3 | IRB Packet Doc 6; restated in manuscript COI declaration |
| Independent Assent Administrator | OSU 1.6 mitigation | IRB Packet Doc 8 |
| Data security plan | OSU 1.7 | IRB Packet Doc 7 |
| Recruitment plan (closed cohort) | OSU 1.5 | IRB Packet Appx B |
| Off-site / multi-state description | OSU 1.9 | IRB Packet Doc 1 §5 |
| Authorship criteria | IEEE 2.2 | Manuscript author list (Ryan + Theodore + Dr. Di); Xiaodong Wang in Acknowledgments only |
| Author contributions section | IEEE biomedical norm 2.8 | Manuscript end-matter |
| AI disclosure | IEEE 2.6 | Manuscript Acknowledgments |
| Data availability | IEEE biomedical norm 2.8 | Manuscript end-matter |

| Requirement | Source | Where it lives in our work |
|---|---|---|
| Adverse event reporting | OSU 1.10 | IRB Packet Appx C |
| Continuing review | OSU 1.10 | IRB Packet Appx C |

# Part 4 — Open items, TBDs, and pre-submission action items

## 4.1 TBDs to be filled in the packet (consolidated)

These are bracketed `[TBD]` placeholders that must have real values written into `IRB_Packet_OSU_sEMG_IMU_Athletic_Practice.md` before Cayuse submission.

| TBD | Where it appears in the packet |
|---|---|
| OSU IRB protocol/approval number | Doc 1 §29; Doc 2A/2B header; Appx A.1; Appx E E1 |
| IAA name, title, department, email, phone | Doc 1 §2; Doc 8 §8.1; Appx D |
| Theodore's exact date of birth (day in September) | Frontmatter; Doc 1 §6; Doc 2B; Doc 3B |
| Ryan's residential court address (city, state, court description) | Doc 1 §5 Site A; Doc 2A "What Will Happen" |
| Theodore's neighborhood fitness center address (Corvallis, OR; facility name if applicable) | Doc 1 §5 Site B; Doc 2B "What Will Happen" |
| Ryan's other parent's name | Doc 2A signature lines |
| Theodore's other parent's name | Doc 2B signature lines |
| OSU HRPP/IRB participant-rights contact (phone + email) | Doc 2A/2B "Questions" section |
| Data retention period [N] years | Doc 1 §17; Doc 7 §7.5 |
| Future-use scope decision | Doc 1 §31; Doc 2A/2B/3A/3B |

| TBD | Where it appears in the packet |
|---|---|
| Cloud-storage decision (OSU-licensed vs. encrypted local) | Doc 7 §7.4 |
| ORCID for each author | Appx E E10 |
| AI-disclosure decision (whether to include and which sections) | Appx A.6; Appx E E8 |
| Conference/journal target | Affects Appx A.3 (data availability) and §2.8 venue norms |

## 4.2 Pre-submission action items

Concrete tasks (verbs with owners). Check each box as completed.

### A. CITI training (all team members must finish before submission)

- ☐ Dr. YanMing Di — CITI Group 1 or 2 (Human Subjects Protection); CITI email matched to Banner email
- ☐ Ryan Wang — CITI Group 1 or 2 via OSU affiliation
- ☐ Theodore (Teddy) Di — CITI Group 1 or 2 via OSU affiliation
- ☐ Xiaodong Wang — CITI Group 1 or 2 via OSU affiliation, or equivalent training accepted by the IRB
- ☐ IAA designee — CITI Group 1 or 2 (Human Subjects Protection)

### B. People and identification

- ☐ Identify and confirm Independent Assent Administrator (OSU faculty/staff; not a parent, co-author, or technical assistant; no publication interest)
- ☐ Confirm Theodore's exact date of birth (day in September) so adult re-consent date in Doc 1 §25 / Doc 2B / Doc 3B is exact
- ☐ Collect Ryan's other parent's full name and relationship for Doc 2A signature lines
- ☐ Collect Theodore's other parent's full name and relationship for Doc 2B signature lines
- ☐ Confirm Dr. Di's OSU Cayuse HE account is active and ONID SSO works

### C. Site descriptions

- ☐ Capture Ryan's residential court address (city + state at minimum) for Doc 1 §5 Site A and Doc 2A
- ☐ Capture Theodore's neighborhood fitness center address (Corvallis, OR; facility name if applicable) for Doc 1 §5 Site B and Doc 2B

- [ ] **Confirm whether the Corvallis neighborhood fitness center requires written site permission / facility authorization for data collection.** If required, obtain a letter of cooperation or document facility approval before Cayuse submission. (Member-access residential or condo gyms typically do not; commercial gyms may.)

## D. Decisions to document

- [ ] Decide data retention period (recommended: 5 years after publication or study closure)
- [ ] Decide future-use scope (recommended: limited to this study and directly derived publications; broader future use needs IRB amendment)
- [ ] Decide cloud-storage vs. encrypted local for Level 2 data (cloud must be OSU-licensed to avoid ISO review)
- [ ] Decide IEEE conference/journal target (affects which §2.8 biomedical norms are mandatory)
- [ ] Decide AI-disclosure inclusion in the manuscript Acknowledgments — include only if AI-assisted drafting was actually used

## E. Conflict-of-interest filings

- [ ] Dr. Di files COI disclosure via OSU Beaver Research Hub disclosing the familial-participant relationship with Theodore Di
- [ ] Dr. Di and Xiaodong Wang sign Doc 6 §6.4 COI Management Plan acknowledgment
- [ ] Dr. Di confirms with OSU COI Office whether familial-participant disclosures require a separate management-plan submission

## F. Re-verifications (do these the week of submission)

- [ ] Re-fetch the URLs in Part 5 and confirm direct quotes still match
- [ ] Update `last_verified` in the frontmatter
- [ ] Verify OSU HRPP/IRB participant-rights contact info from current OSU IRB letterhead
- [ ] Verify OSU consent/assent template version is current (do not reuse pre-Common-Rule forms)
- [ ] Confirm Cayuse HE submission flow with current OSU IRB instructions

## G. Author preparation (for IEEE)

- [ ] Obtain ORCID for Dr. YanMing Di
- [ ] Obtain ORCID for Ryan Wang
- [ ] Obtain ORCID for Theodore (Teddy) Di
- [ ] Confirm IEEE author guide for the chosen target venue (EMBC, JBHI, JTEHM, TBME, or Sensors)

### H. Internal sign-offs (after IRB approval, before data collection starts)

- [ ] Both parents sign Doc 2A parental permission for Ryan
- [ ] Both parents sign Doc 2B parental permission for Theodore (or single-parent justification documented per §46.408(b))
- [ ] IAA conducts/witnesses Ryan's assent privately and signs Doc 3A as Person Obtaining Assent
- [ ] IAA conducts/witnesses Theodore's assent privately and signs Doc 3B as Person Obtaining Assent
- [ ] IAA signs Doc 8 designation acknowledgment

## 4.3 Filled in after submission / approval

- [ ] OSU IRB protocol number (assigned upon Cayuse submission)
- [ ] OSU IRB approval date (assigned upon approval)
- [ ] OSU continuing-review due date (from approval letter)

# Part 5 — Sources and verification URLs

**OSU IRB (Office for Research Integrity):**

- OSU IRB main / submission page
- OSU HRPP Policy and Procedure Manual
- OSU New IRB Applications
- OSU CITI / Education and Advising
- OSU Consent and Assent Templates
- OSU Office of Research Integrity — Conflict of Interest

**IEEE Author Center (publishing ethics):**

- IEEE Submission and Peer Review Policies
- IEEE Ethical Requirements
- IEEE Author Responsibilities
- IEEE Fundamental Publishing Guidelines and Principles
- IEEE PSPB Operations Manual — authoritative manual; section 8.2.1.A (authorship), 8.2.2 (review), 8.2.10 (referencing). Re-verify direct PDF link before submission.

**Reference frameworks:**

- 45 CFR 46 (Common Rule) — `ecfr.gov`

- 45 CFR 46 Subpart D (Children) — `ecfr.gov`
- Declaration of Helsinki (WMA, current revision) — `wma.net`
- ICMJE authorship criteria — `icmje.org`
- CRediT taxonomy — `credit.niso.org`

# Part 6 — Re-verification policy

OSU and IEEE policies change. Before each IRB submission or manuscript submission:

1. Re-fetch the URLs in Part 5 and confirm any direct quotes still match.
2. Update the `last_verified` date in the frontmatter.
3. If a policy changed, log the diff in this file and update affected sections of `IRB_Packet_OSU_sEMG_IMU_Athletic_Practice.md`.

# Part 7 — Protocol-drafting checklist (derived from review)

These are not OSU/IEEE *requirements* per se; they are protocol elements that an internal review of this document flagged as worth including in `IRB_Packet_OSU_sEMG_IMU_Athletic_Practice.md` because they materially strengthen the OSU IRB application or address known reviewer concerns.

## 7.1 Closed-cohort recruitment and no compensation

- No public recruitment; closed N=2 cohort known to study team.
- No compensation, no cost to participants.
- Participation does not affect school standing, sports/athletic opportunities, family relationships, authorship, or other benefits.

## 7.2 Participation–authorship separation

- Authorship and acknowledgment decisions are not contingent on a minor's continued data-collection participation.
- A participant may withdraw from data collection while still receiving appropriate scholarly credit for work already completed, where applicable under publication standards.

- The decision to participate in data collection is documented separately from any manuscript authorship agreement. Refusal or withdrawal from human-subjects participation will **not** be treated as misconduct, failure, or lack of scholarly contribution.

## 7.3 Independent Assent Administrator (IAA) — formal protection

- IAA is **not** a parent, co-author, technical assistant, supervisor, or anyone with a publication interest in the study.
- IAA confirms privately with each minor that participation is voluntary, may be declined or withdrawn without penalty, and is not required to maintain family, athletic, authorship, or academic relationships.

## 7.4 Device safety language (passive-recording only)

- Surface EMG sensors record naturally occurring muscle electrical activity from the skin surface.
- Device is **passive**: no electrical stimulation, no medication, no heat, no therapeutic intervention, battery powered, no diagnostic or treatment claims.
- Participant may remove sensors at any time. Stop on discomfort, skin irritation, dizziness, pain, or request.

## 7.5 No clinical/device-claim framing

- The system is used only for non-clinical research signal recording and engineering analysis.
- Not used for diagnosis, treatment, prevention, monitoring of a medical condition, or clinical decision-making.
- The study does not test a therapeutic or diagnostic device claim. The PI will address any OSU questions about FDA/device applicability if raised, rather than asserting a categorical "not a medical device" determination upfront.
- **No medical feedback to participants.** Participants and parents will not receive medical interpretation, diagnosis, treatment advice, or individualized health recommendations based on the sEMG/IMU data. Athletic coaching, training, or practice decisions remain outside the research protocol.

## 7.6 Ordinary athletic risk vs. research-added risk

- Ordinary athletic activity includes tennis-specific risks for Ryan (court hazards, swing-related strain) and running/fitness-equipment risks for Theodore (treadmill, slip surfaces, equipment use).
- Research does not require participants to exceed normal training intensity.

- Research-added risks are limited to possible mild skin irritation, minor inconvenience, and confidentiality risk.

## 7.7 Off-site / private-site and community-fitness privacy

- No public spectators are invited by the research team.
- No video or audio is recorded.
- No school or team involvement; collection occurs at a private residential court (Ryan, tennis) or a neighborhood fitness center (Theodore, running) only.
- Supervising adult on-site for safety; collection postponed for unsafe weather, court, facility, treadmill, or equipment conditions.
- No data are collected from facility staff, members, patrons, coaches, or any other third party.

## 7.8 Reportable adverse events / deviations (specifics)

- Skin redness lasting beyond a session.
- Rash or blistering from electrodes, tape, or straps.
- Pain or unusual discomfort from device placement.
- Privacy breach, lost device, or lost data.
- Distress or perceived pressure to participate.
- Any athletic injury during a research-recorded session — tennis-related or running/fitness-related, even if likely ordinary sport-related — logged and evaluated.

## 7.9 Adult re-consent operational detail

- New data collection pauses at age 18 until adult informed consent is obtained.
- Pre-18 data remain usable under original parental permission and minor assent unless the now-adult participant withdraws.
- If the now-adult participant declines re-consent, no further data collection for that participant.
- **Withdrawal / data-removal limits.** Participants may request withdrawal of identifiable or coded data to the extent practicable. Data that have already been de-identified, aggregated, analyzed, or included in submitted/published results may not be removable. This limit must be disclosed in the parental permission and minor assent forms.

## 7.10 Data retention and future use

- Retention period: e.g., **5 years after publication or study closure**, identity-key file destroyed earlier when no longer needed (final value TBD by PI).
- De-identified aggregate summary metrics may be retained longer if IRB approves.

- Future use: data are used only for the approved study and related publications/presentations described in this IRB application; materially different future use requires IRB amendment.

## 7.11 Manuscript ethics statement (IEEE-ready)

> The study was reviewed and approved by the Oregon State University Human Research Protection Program / Institutional Review Board, protocol #[TBD]. Written parental permission and written minor assent were obtained before participation. The study was conducted in accordance with the Declaration of Helsinki.

## 7.12 Manuscript COI declaration (IEEE-ready)

> The authors disclose that one minor participant/co-author is the child of the principal investigator and corresponding author, and the second minor participant/co-author is the child of a technical contributor named in the Acknowledgments. To mitigate potential undue influence, minor assent was obtained through an independent assent administrator, and parental permission was documented before participation. The authors have no financial conflicts of interest.

## 7.13 Manuscript AI-disclosure (IEEE-ready, only if true)

> AI-assisted drafting tools were used to assist with organization and language editing of the manuscript. All study design, data collection, analysis, interpretation, and final manuscript content were reviewed and approved by the authors.